ANOVA In-Class Activity

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Overview of Steps (from slide)

1. Check for normality
2. Check homogeneity of variances → **Factorial / Independent ANOVA**
   Check sphericity of variances → **Repeated Measures ANOVA**
3. Choose the appropriate test
4. Only if F is significant → run post-hoc test
5. Report effect size
6. Plot the data

Q1: Memory Scores

Do children with neurodevelopmental disorders have lower memory scores?
Design: **One-Way Independent ANOVA** (different participants per group)

Data

```
regular  <- c(24, 22, 19, 22, 28, 26, 28, 24, 30, 29, 25, 20, 17, 19, 18, 26, 2
7, 24, 27, 27)
autism   <- c(15, 2, 1, 21, 3, 10, 9, 8, 3, 7, 6, 18, 2, 5, 2, 5, 0, 27)
epilepsy <- c(30, 15, 34, 26, 14, 28, 17, 29, 25, 11, 37, 36, 34, 22, 18, 5, 12,
10, 15)

# Combine into a data frame
score <- c(regular, autism, epilepsy)
group <- factor(rep(c("Regular", "Autism", "Epilepsy"),
                    times = c(length(regular), length(autism), length(epileps
y))))
df <- data.frame(score, group)

# Descriptive statistics per group
tapply(df$score, df$group, function(x) round(c(n=length(x), mean=mean(x), sd=sd
(x)), 2))
```

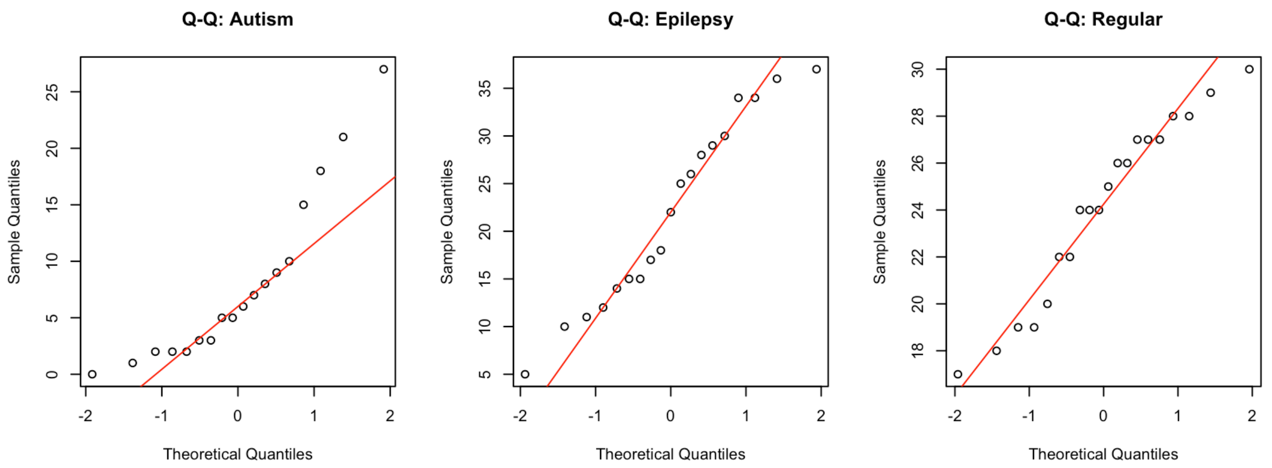
```
## $Autism
##      n  mean    sd
## 18.00  8.00  7.58
##
## $Epilepsy
##      n  mean    sd
## 19.00 22.00  9.83
##
## $Regular
##      n  mean    sd
## 20.00 24.10  3.89
```

Step 1: Check Normality (Shapiro-Wilk + Q-Q Plots)

```
# Shapiro-Wilk test per group
for (g in levels(df$group)) {
  x <- df$score[df$group == g]
  sw <- shapiro.test(x)
  cat(g, ": W =", round(sw$statistic, 3), ", p =", round(sw$p.value, 3),
      ifelse(sw$p.value > 0.05, "-> Normal", "-> NOT Normal"), "\n")
}
```

```
## Autism : W = 0.856 , p = 0.011 -> NOT Normal
## Epilepsy : W = 0.947 , p = 0.347 -> Normal
## Regular : W = 0.942 , p = 0.262 -> Normal
```

```
# Q-Q plots
par(mfrow = c(1, 3))
for (g in levels(df$group)) {
  x <- df$score[df$group == g]
  qqnorm(x, main = paste("Q-Q:", g))
  qqline(x, col = "red")
}
```




```
par(mfrow = c(1, 1))
```

Step 2: Check Homogeneity of Variances

```
# Bartlett's test (assumes normality)
bartlett.test(score ~ group, data = df)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  score by group
## Bartlett's K-squared = 13.889, df = 2, p-value = 0.0009641
```

```
# Levene's test (robust to non-normality) - manual, no extra package
levene_manual <- function(y, group) {
  group_medians <- tapply(y, group, median)      # median-based for robustness
  z <- abs(y - group_medians[group])             # absolute deviations
  anova(lm(z ~ group))
}
levene_manual(df$score, df$group)
```

```
## Analysis of Variance Table
##
## Response: z
##           Df  Sum Sq Mean Sq F value   Pr(>F)
## group      2  277.93 138.967  7.4901 0.001346 **
## Residuals 54 1001.88  18.553
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> If p > .05: variances are homogeneous → proceed with standard ANOVA

Step 3: One-Way ANOVA

```
model <- aov(score ~ group, data = df)
summary(model)
```

```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## group        2   2842  1421.3   25.53 1.57e-08 ***
## Residuals   54   3006    55.7
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 4: Effect Size (η² — Eta-squared)

```
# eta-squared = SS_between / SS_total
ss      <- summary(model)[[1]][["Sum Sq"]]
eta_sq  <- ss[1] / sum(ss)
cat("Eta-squared (eta^2) =", round(eta_sq, 3), "\n")
```

```
## Eta-squared (eta^2) = 0.486
```

```
cat("Benchmarks: small = 0.01, medium = 0.06, large = 0.14\n")
```

```
## Benchmarks: small = 0.01, medium = 0.06, large = 0.14
```

Step 5: Post-Hoc Test (Tukey HSD — only if F significant)

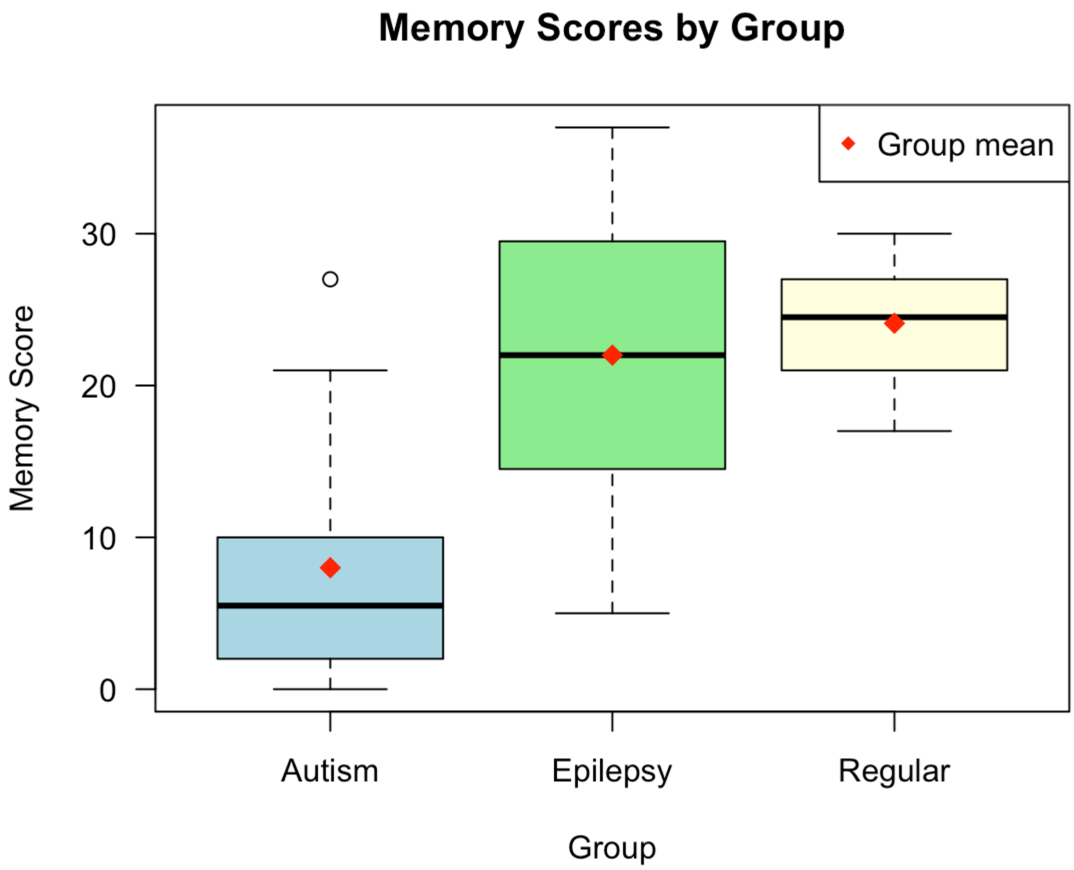
```
TukeyHSD(model)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = score ~ group, data = df)
##
## $group
##                    diff       lwr       upr     p adj
## Epilepsy-Autism  14.0  8.085948 19.914052 0.0000015
## Regular-Autism   16.1 10.258319 21.941681 0.0000000
## Regular-Epilepsy  2.1 -3.660203  7.860203 0.6560029
```

Step 6: Plot

```
boxplot(score ~ group, data = df,
        col  = c("lightblue", "lightgreen", "lightyellow"),
        main = "Memory Scores by Group",
        xlab = "Group", ylab = "Memory Score", las = 1)

# Mark group means
means <- tapply(df$score, df$group, mean)
points(1:length(means), means, pch = 18, col = "red", cex = 1.5)
legend("topright", legend = "Group mean", pch = 18, col = "red")
```



Q2: Driving Ability

What extent of sleep deprivation affects driving ability?

```
night1 <- c(15, 18, 20, 15, 12, 18, 16, 17, 14, 19, 20, 15, 16, 18, 19, 15, 17,
18, 19, 15)
night2 <- c(10, 16, 13, 11,  9, 14, 13, 14, 15, 14, 12, 13, 14, 12, 11, 15, 14,
16, 11, 14)
night3 <- c( 5,  3,  9,  6,  4,  7,  8,  2,  4,  6,  9,  5,  3,  7,  1,  8,  7,
3,  7,  3)
```

Condition 1: Same participants in all 3 groups → Repeated Measures ANOVA

Step 1: Normality

```
data_list <- list("1 Night" = night1, "2 Nights" = night2, "3 Nights" = night3)
for (nm in names(data_list)) {
  sw <- shapiro.test(data_list[[nm]])
  cat(nm, ": W =", round(sw$statistic, 3), ", p =", round(sw$p.value, 3),
      ifelse(sw$p.value > 0.05, "-> Normal", "-> NOT Normal"), "\n")
}
```

```
## 1 Night : W = 0.945 , p = 0.295 -> Normal
## 2 Nights : W = 0.949 , p = 0.345 -> Normal
## 3 Nights : W = 0.948 , p = 0.333 -> Normal
```

Step 2: Sphericity (Mauchly's Test)

```
# mauchly.test() is built into base R
Y     <- cbind(night1, night2, night3)
idata <- data.frame(deprivation = factor(c("1_Night", "2_Nights", "3_Nights")))
fit   <- lm(Y ~ 1)
mauchly.test(fit, X = ~ 1, idata = idata)
```

```
##
##  Mauchly's test of sphericity
##  Contrasts orthogonal to
##  ~1
##
##
## data:  SSD matrix from lm(formula = Y ~ 1)
## W = 0.93042, p-value = 0.5225
```

> If p > .05: sphericity holds → standard RM-ANOVA F values are valid
> If p < .05: sphericity violated → apply Greenhouse-Geisser correction (epsilon correction on df)

Step 3: Repeated Measures ANOVA

```
n     <- length(night1)
df_rm <- data.frame(
  subject     = factor(rep(1:n, 3)),
  deprivation = factor(rep(c("1_Night","2_Nights","3_Nights"), each = n),
                       levels = c("1_Night","2_Nights","3_Nights")),
  score       = c(night1, night2, night3)
)

# Error(subject/deprivation) partitions within-subject variance
rm_model <- aov(score ~ deprivation + Error(subject/deprivation), data = df_rm)
summary(rm_model)
```

```
##
## Error: subject
##           Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 19  109.7   5.775
##
## Error: subject:deprivation
##             Df Sum Sq Mean Sq F value Pr(>F)
## deprivation  2   1363   681.5   162.9 <2e-16 ***
## Residuals   38    159     4.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 4: Effect Size (Partial η²)

```
rm_sum <- summary(rm_model)
ss_dep <- rm_sum$`Error: subject:deprivation`[[1]]["deprivation", "Sum Sq"]
ss_res <- rm_sum$`Error: subject:deprivation`[[1]]["Residuals",  "Sum Sq"]
eta_rm <- ss_dep / (ss_dep + ss_res)
cat("Partial Eta-squared =", round(eta_rm, 3), "\n")
```

```
## Partial Eta-squared = 0.896
```

Step 5: Post-Hoc (Bonferroni-corrected paired t-tests)

```
pairwise.t.test(df_rm$score, df_rm$deprivation,
                paired          = TRUE,
                p.adjust.method = "bonferroni")
```

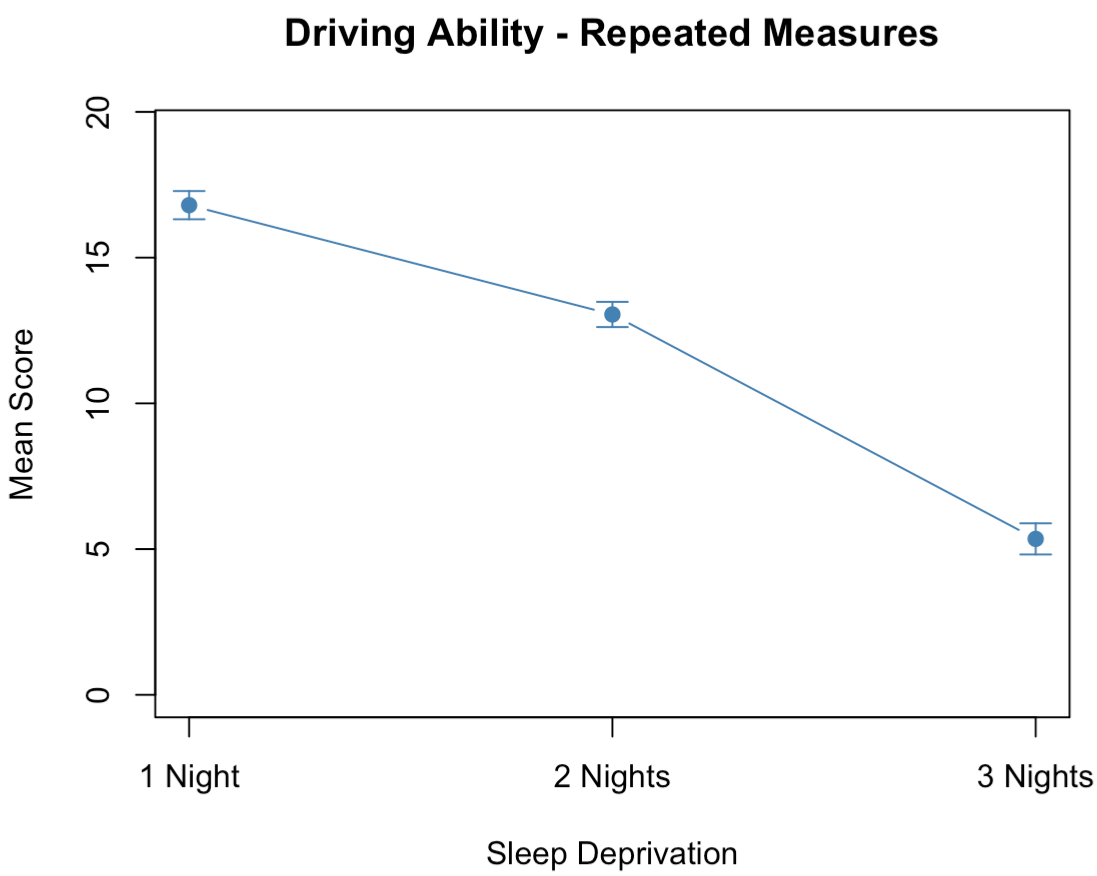
```
##
##  Pairwise comparisons using paired t tests
##
## data:  df_rm$score and df_rm$deprivation
##
##          1_Night 2_Nights
## 2_Nights 1.2e-05 -
## 3_Nights 3.8e-13 6.0e-09
##
## P value adjustment method: bonferroni
```

Step 6: Plot

```r
means_rm <- tapply(df_rm$score, df_rm$deprivation, mean)
se_rm    <- tapply(df_rm$score, df_rm$deprivation,
                   function(x) sd(x) / sqrt(length(x)))

plot(1:3, means_rm, type = "b", pch = 19, col = "steelblue",
     ylim = c(0, max(means_rm + se_rm) + 2),
     xaxt = "n",
     main = "Driving Ability - Repeated Measures",
     xlab = "Sleep Deprivation", ylab = "Mean Score")
axis(1, at = 1:3, labels = c("1 Night","2 Nights","3 Nights"))

# Error bars (±1 SE)
arrows(1:3, means_rm - se_rm, 1:3, means_rm + se_rm,
       angle = 90, code = 3, length = 0.08, col = "steelblue")
```



Condition 2: Different participants in all 3 groups → Independent One-Way ANOVA

Step 1: Normality

```
for (nm in names(data_list)) {
  sw <- shapiro.test(data_list[[nm]])
  cat(nm, ": W =", round(sw$statistic, 3), ", p =", round(sw$p.value, 3),
      ifelse(sw$p.value > 0.05, "-> Normal", "-> NOT Normal"), "\n")
}
```

```
## 1 Night : W = 0.945 , p = 0.295 -> Normal
## 2 Nights : W = 0.949 , p = 0.345 -> Normal
## 3 Nights : W = 0.948 , p = 0.333 -> Normal
```

Step 2: Homogeneity of Variances

```
score_ind <- c(night1, night2, night3)
cond_ind  <- factor(rep(c("1_Night","2_Nights","3_Nights"), each = 20),
                    levels = c("1_Night","2_Nights","3_Nights"))

bartlett.test(score_ind ~ cond_ind)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  score_ind by cond_ind
## Bartlett's K-squared = 0.83327, df = 2, p-value = 0.6593
```

```
levene_manual(score_ind, cond_ind)
```

```
## Analysis of Variance Table
##
## Response: z
##           Df Sum Sq Mean Sq F value Pr(>F)
## group      2    2.5  1.2500  0.9241 0.4027
## Residuals 57   77.1  1.3526
```

Step 3: One-Way ANOVA

```
model_ind <- aov(score_ind ~ cond_ind)
summary(model_ind)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## cond_ind     2 1363.0   681.5   144.6 <2e-16 ***
## Residuals   57  268.7     4.7
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 4: Effect Size (η²)

```
ss_ind  <- summary(model_ind)[[1]][["Sum Sq"]]
eta_ind <- ss_ind[1] / sum(ss_ind)
cat("Eta-squared =", round(eta_ind, 3), "\n")
```

```
## Eta-squared = 0.835
```

Step 5: Post-Hoc (Tukey HSD)

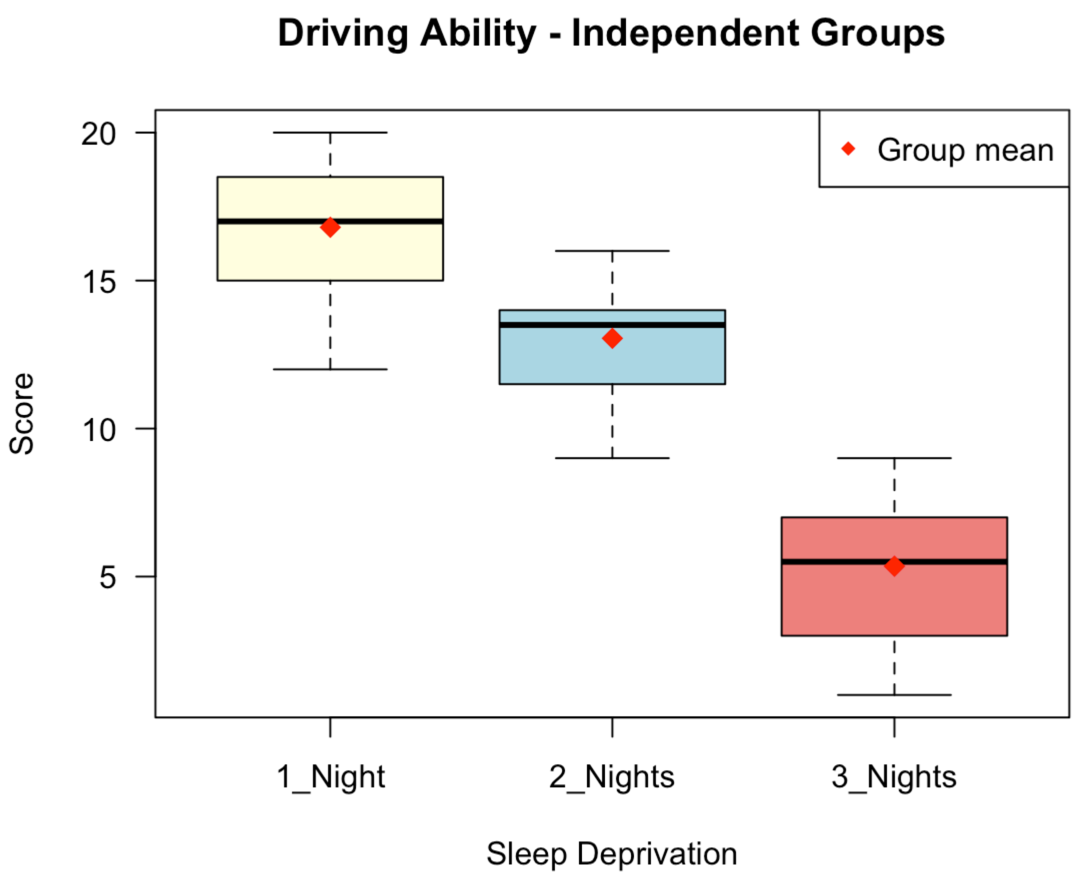
```
TukeyHSD(model_ind)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = score_ind ~ cond_ind)
##
## $cond_ind
##                      diff        lwr       upr   p adj
## 2_Nights-1_Night   -3.75  -5.402219 -2.097781 3.2e-06
## 3_Nights-1_Night  -11.45 -13.102219 -9.797781 0.0e+00
## 3_Nights-2_Nights  -7.70  -9.352219 -6.047781 0.0e+00
```

Step 6: Plot

```
boxplot(score_ind ~ cond_ind,
        col  = c("lightyellow", "lightblue", "lightcoral"),
        main = "Driving Ability - Independent Groups",
        xlab = "Sleep Deprivation", ylab = "Score", las = 1)

means_ind <- tapply(score_ind, cond_ind, mean)
points(1:3, means_ind, pch = 18, col = "red", cex = 1.5)
legend("topright", legend = "Group mean", pch = 18, col = "red")
```



Q3: Exam Performance

Does school type affect exam scores?
Design: **One-Way Independent ANOVA** (different participants per group)

Data

```
home_school <- c(89, 75, 49, 87, 84, 68, 88, 78, 77, 93, 67, 79, 69, 88, 91)
boarding    <- c(85, 78, 59, 77, 63, 88, 71, 73, 69, 80, 72, 68, 66, 59, 70)
regular_day <- c(91, 88, 84, 81, 91, 75, 69, 93, 95, 85, 87, 84, 83, 80, 77)

score_exam <- c(home_school, boarding, regular_day)
school     <- factor(rep(c("Home","Boarding","Regular Day"), each = 15),
                     levels = c("Home","Boarding","Regular Day"))
df_exam    <- data.frame(score_exam, school)

tapply(df_exam$score_exam, df_exam$school,
       function(x) round(c(n=length(x), mean=mean(x), sd=sd(x)), 2))
```

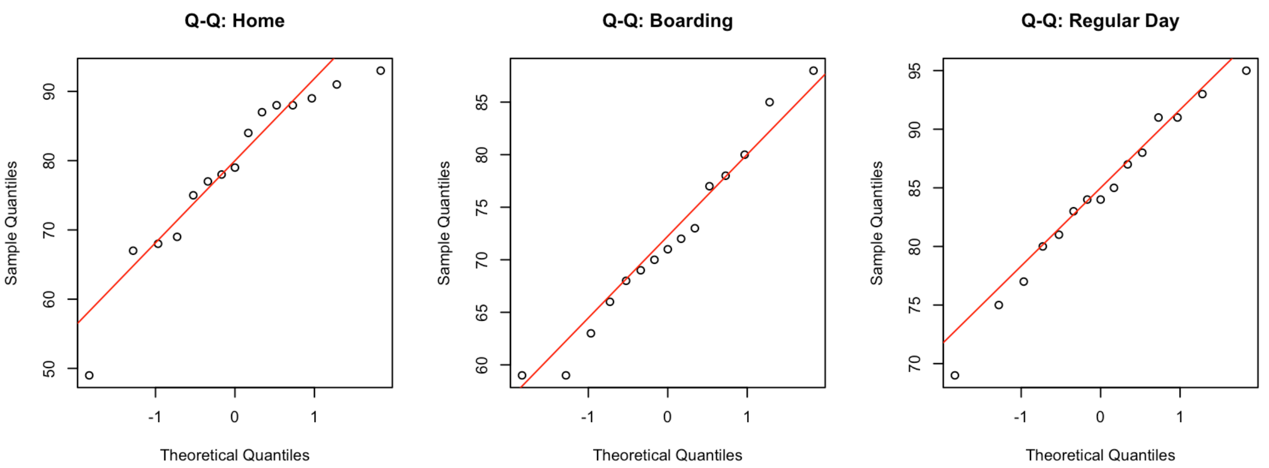
```
## $Home
##     n  mean    sd
## 15.00 78.80 11.88
##
## $Boarding
##     n  mean    sd
## 15.00 71.87  8.60
##
## $`Regular Day`
##    n mean   sd
## 15.0 84.2  7.1
```

Step 1: Normality

```
for (g in levels(df_exam$school)) {
  x  <- df_exam$score_exam[df_exam$school == g]
  sw <- shapiro.test(x)
  cat(g, ": W =", round(sw$statistic, 3), ", p =", round(sw$p.value, 3),
      ifelse(sw$p.value > 0.05, "-> Normal", "-> NOT Normal"), "\n")
}
```

```
## Home : W = 0.905 , p = 0.115 -> Normal
## Boarding : W = 0.969 , p = 0.847 -> Normal
## Regular Day : W = 0.975 , p = 0.923 -> Normal
```

```
par(mfrow = c(1, 3))
for (g in levels(df_exam$school)) {
  x <- df_exam$score_exam[df_exam$school == g]
  qqnorm(x, main = paste("Q-Q:", g))
  qqline(x, col = "red")
}
```




```
par(mfrow = c(1, 1))
```

Step 2: Homogeneity of Variances

```
bartlett.test(score_exam ~ school, data = df_exam)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  score_exam by school
## Bartlett's K-squared = 3.7361, df = 2, p-value = 0.1544
```

```
levene_manual(df_exam$score_exam, df_exam$school)
```

```
## Analysis of Variance Table
##
## Response: z
##           Df  Sum Sq Mean Sq F value Pr(>F)
## group      2  108.98  54.489  1.6477 0.2047
## Residuals 42 1388.93  33.070
```

Step 3: One-Way ANOVA

```
model_exam <- aov(score_exam ~ school, data = df_exam)
summary(model_exam)
```

```
##             Df Sum Sq Mean Sq F value  Pr(>F)
## school       2   1147   573.4   6.476 0.00354 **
## Residuals   42   3719    88.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 4: Effect Size (η²)

```
ss_exam  <- summary(model_exam)[[1]][["Sum Sq"]]
eta_exam <- ss_exam[1] / sum(ss_exam)
cat("Eta-squared =", round(eta_exam, 3), "\n")
```

```
## Eta-squared = 0.236
```

```
cat("Benchmarks: small = 0.01, medium = 0.06, large = 0.14\n")
```

```
## Benchmarks: small = 0.01, medium = 0.06, large = 0.14
```

Step 5: Post-Hoc (Tukey HSD)

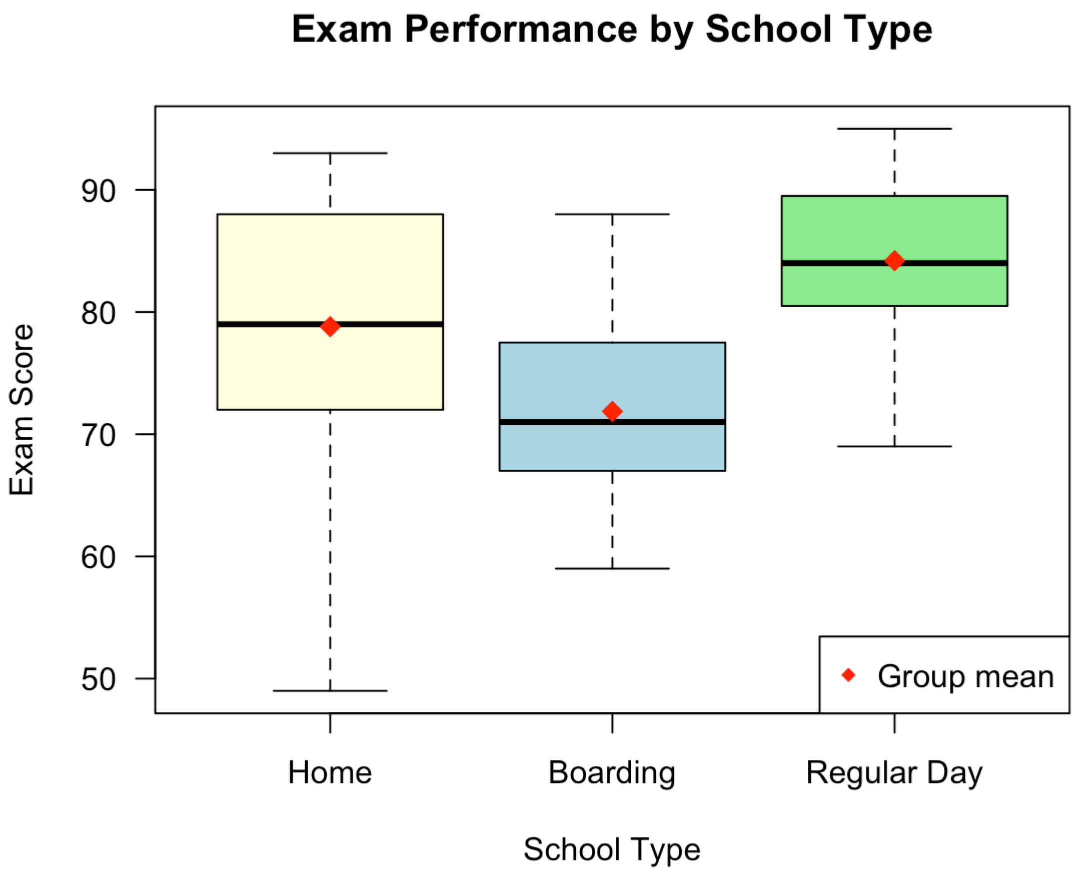
```
TukeyHSD(model_exam)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = score_exam ~ school, data = df_exam)
##
## $school
##                          diff        lwr       upr     p adj
## Boarding-Home        -6.933333 -15.280640  1.413973 0.1203553
## Regular Day-Home      5.400000  -2.947306 13.747306 0.2689648
## Regular Day-Boarding 12.333333   3.986027 20.680640 0.0024280
```

Step 6: Plot

```
boxplot(score_exam ~ school, data = df_exam,
        col  = c("lightyellow", "lightblue", "lightgreen"),
        main = "Exam Performance by School Type",
        xlab = "School Type", ylab = "Exam Score", las = 1)

means_exam <- tapply(df_exam$score_exam, df_exam$school, mean)
points(1:3, means_exam, pch = 18, col = "red", cex = 1.5)
legend("bottomright", legend = "Group mean", pch = 18, col = "red")
```



Summary

Question	Design	Normality	Variance Check	Test	Post-Hoc
Q1 Memory	Independent	`shapiro.test`	`bartlett.test` / `levene_manual`	`aov`	`TukeyHSD`
Q2 Cond.1 Driving	Repeated Measures	`shapiro.test`	`mauchly.test` (sphericity)	`aov + Error()`	`pairwise.t.test` (Bonferroni)
Q2 Cond.2 Driving	Independent	`shapiro.test`	`bartlett.test` / `levene_manual`	`aov`	`TukeyHSD`
Q3 Exam	Independent	`shapiro.test`	`bartlett.test` / `levene_manual`	`aov`	`TukeyHSD`

All code uses base R only — no extra packages required.